

A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

prototype/ecm · runs in log/okuda_ECM · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets $\times$ fields $\times$ operators $\times$ schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity / set	representation	principal operators
epithelium vertex, cell	3D active vertex model	seed_mesh_3d build the shell cell_geometry_3d area, perimeter, volume per cell shape_energy_3d the mechanics morphogen_growth_3d grow each cell's target divide_3d divide on volume doubling ecm_growth_gate_3d matrix stress slows the cycle [7]
junction a relation, not a set	half-edges of the mesh, state keyed by vertex pair	junction_myosin per-junction myosin reconnect_t1_3d neighbour exchange (T1)
basement membrane basement_membrane_particle	nodes + explicit crosslinks	seed_basement_membrane lay the sheet down basement_membrane_bond the crosslink force basement_membrane_remodel rest lengths creep basement_membrane_bond_break fragmentation basement_membrane_secrete add material as it grows integrin_adhesion tether to the surface
surface surface	one node per membrane node: direction, radius	surface_track write the recorded boundary
interstitial ECM mpm_particle	MPM fibres	seed_ecm fibres, alignment, density cell_to_ecm the epithelium's push cell_exclude_3d non-penetration backstop ecm_stress colour by von Mises stress
(field) mpm_grid	MLS-MPM background grid	mpm_scatter / mpm_grid_update / mpm_gather the solve

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f 's volume is the wedge from the tissue centre, $v_f = \frac{1}{3} \mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E = \sum_f \left[K_A (A_f - A_f^0)^2 + K_P (P_f - P_f^0)^2 + \frac{1}{2} \Gamma P_f^2 + K_V (v_f - v_f^0)^2 \right] + \Lambda \sum_e m_e \ell_e + K_R \sum_i (|\mathbf{x}_i| - R_0)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta \mu \nabla E$ with a per-step cap. The factor m_e is new (§6); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f ,

$$s_f \leftarrow s_f (1 + \lambda(\rho + \text{Hill}(a_f))), \quad A_f^0 = A_f^{0,\text{init}} s_f^2, \quad P_f^0 = P_f^{0,\text{init}} s_f, \quad v_f^0 = v_f^{0,\text{init}} s_f^3, \quad (2)$$

and **divide_3d** splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \rightarrow 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is a material, not a set of springs, so it is solved the way materials are: as a cloud of points that carry the material and a background grid that does the physics. Each frame the points hand their mass and momentum to the grid nodes nearest them, the grid resolves everything arriving at each node into one velocity, and the points read that velocity back and move. Nothing is ever computed between two points directly — which is why bodies that never touch can still push on one another, and why the grid spacing sets the finest thing the matrix can feel.

We seed it as fibres rather than a uniform block, because a fibrous matrix and a homogeneous one respond differently to being pushed, and the interesting behaviour — splaying, dragging, a stress front travelling outward — lives in the fibres.

4. The matrix on its own: a block, dropped

Every run above loads the matrix the same way — slowly, outward, everywhere at once — so none of them says whether it carries stress the way a fibrous material should. A block of it falling under gravity onto the floor is the cheapest load that *arrives*. Four runs (02b, 02c, 02d_drag2, 02d_drag8), each 90,000 particles on 1500 strands of 60, 720 frames at $\Delta t = 3.2 \times 10^{-3}$, differing in one number apiece.

What the grid can resolve, before anything falls. A grid cell is $1/64 = 0.0156$ box units. A strand is 0.09 long, so 5.8 cells; its particles sit 0.0015 apart, a tenth of a cell; and the across-strand jitter that gives it thickness is 0.004, a quarter of a cell. The grid therefore resolves a strand along its length and *cannot* resolve it across — about ten particles of one strand land in a single cell and are interpolated into one velocity — and no force is ever computed between two particles of the same strand. The fibres are an arrangement of the mass, not a mechanism.

The strands do nothing a continuum would not. Through an impact that compresses the block to 0.914 of its seeded height, a strand’s end-to-end length falls to 0.982 of the 0.0896 box units it was seeded with and returns to 0.999; its bow — the RMS distance of its own particles from its own end-to-end axis — goes $0.00824 \rightarrow 0.00827 \rightarrow 0.00826$ box units, an excursion of 0.4%. Nothing buckles, nothing stays stretched, and the orientation order parameter about the drop axis moves by 0.04. This is the prediction of the constitutive law, which is isotropic fixed-corotated and reads the strand direction nowhere, and the measurement agrees with it.

The readout had been the wrong quantity. `ecm_stress` defaulted to $|J - 1|$, the local volume change, and an impact is a shear. With `store_stress: true` on `mpm_scatter` the solver keeps the Cauchy stress it already computed to make the grid forces, and `measure: vonmises` reads its invariant instead. The two readings differ by a factor of 250 on the same trajectory — at frame 200, $|J - 1|$ gives mean 0.0040 and p99 0.0173, the von Mises gives mean 1.11 and p99 4.16 in stress units, peaking at p99 42.4 just after first contact. The positions are bit-identical between the two runs, so this is a readout and not a mechanism; it costs 45% more wall clock ($12.5 \rightarrow 18.1$ minutes) and it is the difference between seeing the front and not.

The block rings, and an extracellular matrix does not. After the impact every shape metric oscillates: the measured period is 24.2 frames, 0.0775 s, 12.9 Hz, against the 17.1 Hz a free-free bar of length 0.32 at $c = \sqrt{E/\rho} = 10.95$ box units per second would have, so it is the block’s own elastic breathing mode and not numerical noise. It is nonetheless biologically wrong. The stock `drag` of 0.05 is a velocity decay time of $1/\text{drag} = 20$ s — 260 ringing periods, and 8.7 times the whole 2.31 s run — so this matrix is effectively undamped, whereas a hydrated matrix dissipates by solvent forced through the network and at $\text{Re} \sim 10^{-10}$ cannot ring at all. `drag` is the right lever for that and needs no new operator: it is a Stokes term on the network velocity, which is what Darcy drag is.

run	<code>drag</code>	ripple, as a fraction of the seeded height	height returned at the first impact	bounding volume at the last frame	median residual displacement (box units)
02c	0.05	0.0081	0.707	1.172	0.0360
02d_drag2	2	0.0048	0.293	0.999	0.0026
02d_drag8	8	0.00054	0.048	0.997	0.0011

Two things follow, and the second was invisible until the first was fixed. The damping that removes the ringing removes the bounce with it — at `drag` = 8 the block returns 4.8% of what it fell, so there is no bouncing experiment in the regime where the material is physical, and the three impacts of 02b are a property of an undamped solid rather than of a matrix. And the slow swelling that 02b/02c showed — bounding volume rising to 1.34 of its seeded value by frame 560, the median particle displacement with the block’s own translation removed reaching 0.036 box units, which is 2.3 grid cells — *was not the material*. At `drag` = 2 it is gone: volume 0.999, residual displacement 0.0026, a factor of 13.6 smaller. It was undamped particle noise accumulating, and it had been read as creep.

What would actually discriminate a fibre model. None of the above needs fibres to explain it, so none of it can falsify a fibre model either. Three loadings would, in increasing cost. Compress the block between plates to 20% and release: the tangent modulus at 5% against 20% strain is the stiffening ratio, which a network of straightening fibres raises by one to two orders of magnitude [9, 12] and an isotropic solid does not raise at all; the residual strain after unloading is the plasticity; and the effective Poisson ratio distinguishes a solid that bulges from a network that densifies. Pull the same block by 10% and push it by 10%: the ratio of secant moduli is 1 by construction for the present law and must exceed 3 for anything that buckles in compression [10]. And contract a sphere at the centre by 10% and fit the radial decay of displacement $u(r) \propto r^{-n}$: an incompressible linear continuum gives $n = 2$, a fibrous matrix n between 0.5 and 1 [11], and that single exponent is the property on which the whole spheroid–matrix coupling of §5 rests.

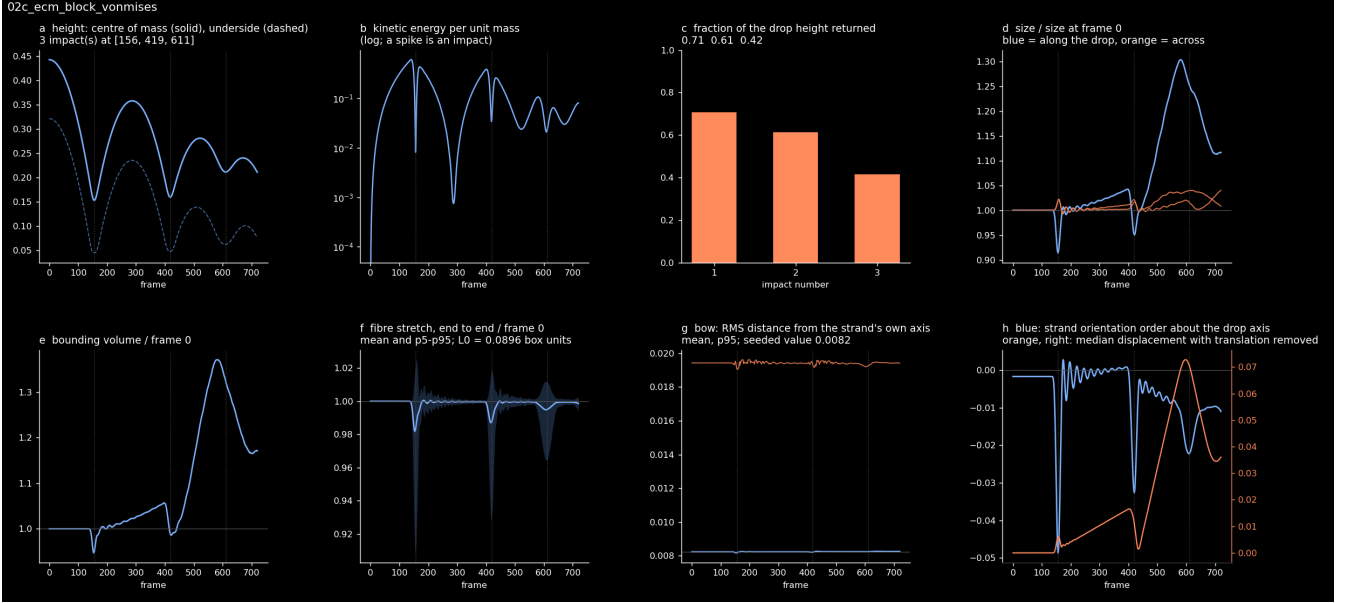


Figure 1: 02c, the undamped block, 721 frames. **a–c** the bulk: three impacts, and the height returned falling $0.707 \rightarrow 0.612 \rightarrow 0.416$ — which reads as dissipation and is in fact the block coming apart, since **e** shows its bounding volume growing to 1.34 of the seeded value. **d–e** ring at 12.9 Hz, the block’s own elastic mode. **f–g** are the fibre state and are the point: end-to-end stretch recovers to 0.999 and the bow moves by 0.4%, so the strands ride the continuum. Everything in **e** and the orange trace in **h** is removed by **drag**= 2.

5. Spheroid $\leftrightarrow$ ECM

The two solvers cannot share a world: the matrix grid is fixed to the unit cube, the epithelium lives in a box fifty times larger, and shrinking the tissue to fit changes which term of its energy dominates and stops it growing. So they are run as two passes joined by one geometric scale, and what crosses between them is a *recorded shape* rather than a live tissue.

That recording needs somewhere to live, and this is what the **surface** Level is. It is a set of nodes, one per membrane node, each holding a fixed direction from the tissue centre and the radius the epithelium reached in that direction at that moment. It is not a mesh and carries no mechanics; it is the epithelium’s outer boundary, written down. Before it existed the same information was a lookup table, which meant nothing could act on it and nothing could be attached to it — promoting it to a Level is what lets other things bind to the tissue surface as an object rather than consult it as data.

Everything outside the epithelium then reads that one surface. Matrix fibres that end up inside it are pushed back out, first softly and then, as a backstop, by being placed outside:

$$\mathbf{a}_i = k_c (R(\hat{\mathbf{u}}_i, t) - r_i)_+ \hat{\mathbf{u}}_i, \quad (3)$$

and the force the fibres push back with is collected by direction and divided by the area each direction covers, giving a pressure over the surface. A cell facing a stressed matrix then grows more slowly, which is what a spheroid grown under a measured external stress does [7, 8]:

$$g = f + \frac{1 - f}{1 + (P/p_{1/2})^n}, \quad s_{\text{now}} \leftarrow s_{\text{prev}} (1 + g(f_{\text{grow}} - 1)). \quad (4)$$

Because it is the growth *rate* that is slowed and not the size directly, a small difference compounds over hundreds of frames — which is how a modest difference in stress between one direction and another can become a difference in shape.

The coupling still runs one way. The surface is a recording, so it drives the matrix and the membrane and neither reaches back, and the second pass cannot change the tissue it is reacting to. Making it mutual means letting an operator write to **surface** instead of only reading it, which is now a change to one Level rather than a change to the architecture.

6. The basement membrane

A basement membrane is a thin sheet of protein an epithelium lays down underneath itself and sits on. It has to do two things at once that pull against each other: hold together as a sheet, and keep covering a surface that grows.

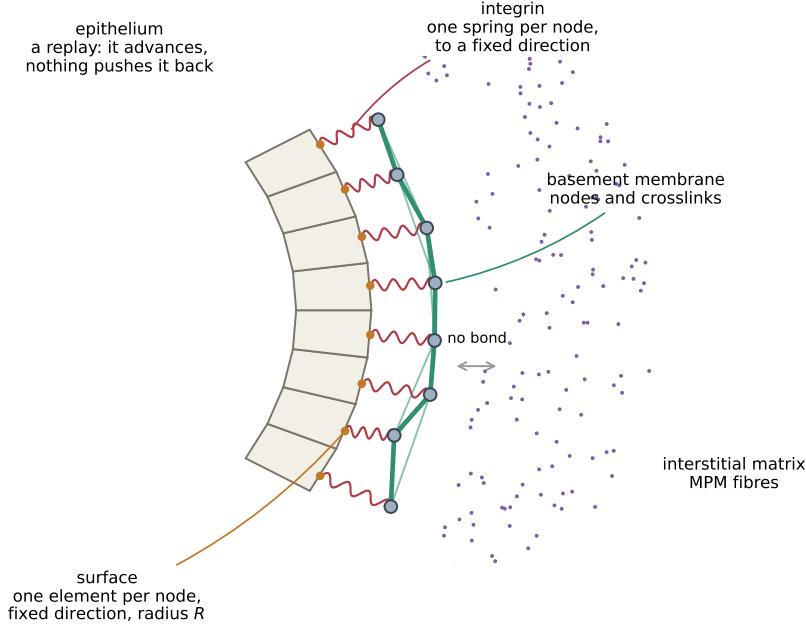


Figure 2: Four objects, and the arrows between them are the model. The epithelium advances on a recorded schedule and nothing pushes it back. Its outer boundary is the **surface** Level: one node per membrane node, each holding a fixed direction and the radius the tissue reached in it. The membrane is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside, each node tied to its own surface node by one red *tether*. The matrix beyond is never bonded to.

We model the load-bearing part of it — the crosslinked collagen IV network — as nodes joined by springs.

Three rules. A crosslink is a spring that slowly *forgets* being stretched and snaps if stretched too far; a node is tied to a fixed *direction* on the surface, which is what makes it adhesion rather than contact, since a node free to slide to the nearest point would never be stretched at all; and everything moves at a speed set by force against drag, with no mass:

$$\gamma \dot{\mathbf{x}}_i = \sum_{j \sim i} \mathbf{f}_{ij} + \mathbf{f}_i^{\text{adh}}, \quad \mathbf{f}_{ij} = k_b (L_{ij} - \ell_{ij}^0) \hat{\mathbf{d}}_{ij}, \quad \dot{\ell}_{ij}^0 = (L_{ij} - \ell_{ij}^0) / \tau_r, \quad (5)$$

$$\mathbf{f}_i^{\text{adh}} = \kappa (\mathbf{a}_i - \mathbf{x}_i), \quad \mathbf{a}_i = \mathbf{c} + \hat{\mathbf{u}}_i (R(\hat{\mathbf{u}}_i, t) + \delta), \quad \hat{\mathbf{u}}_i \text{ frozen at seeding}, \quad (6)$$

with a crosslink dying once $(L - \ell^0) / \ell^0 > \varepsilon_b$.

Forgetting is not optional: a sheet that only stretches cannot enclose a surface that more than triples in size. And because area grows as the square of the radius while new material is added as a fraction of what is already there, the sheet must be *secreted* as the tissue grows or it thins.

Two consequences of having no mass. Only the ratio k_b / γ appears in Eq. (5) — a relaxation *rate* — so a stiffness on its own has no meaning here, and the question worth asking is whether the sheet relaxes faster than the tissue grows. The same ratio bounds what a fixed time step can carry, $\Delta t (z k_b + \kappa) / \gamma < 2$ for z crosslinks meeting at a node, which is about twenty times the bound an inertial sheet would have. Raising γ moves it further, at the price of a sheet that responds more slowly.

Finally, what is called adhesion here is a tether to a surface, not a receptor: the dominant anchor in a real epithelium is to laminin, which is not modelled, and there is no attachment to the surrounding stroma either, so this tether is the only thing holding the sheet.

7. Junction myosin

In the vertex model every cell–cell contact pulls with the same strength. Real contractility is not uniform: myosin accumulates on some junctions and not others, and where it accumulates the junction pulls harder and shortens, which is how cells exchange neighbours. Giving each junction its own myosin is what lets that happen.

Each junction carries an amount of myosin m_e multiplying the line tension, chasing a target on its own timescale:

$$\Lambda \sum_e \ell_e \longrightarrow \Lambda \sum_e m_e \ell_e, \quad m_e^{ss} = a \left(1 + \beta (x_e / \langle x \rangle - 1) \right), \quad \dot{m}_e = (m_e^{ss} - m_e) / \tau_m. \quad (7)$$



Figure 3: Two cells share an edge, and that edge *is* the junction — it belongs to neither, which is why its state is keyed by the pair of vertices rather than stored on a cell. Myosin sits on the shared edge and multiplies its tension. Right: the feedback, which is the only thing here the original model cannot already do.

What matters is what x_e is. If it is the junction's *length*, the loop reads *longer* $\rightarrow$ *more myosin* $\rightarrow$ *shorter*, which evens the cells out and suppresses neighbour exchange. Myosin is in fact recruited by *tension* [6], and accumulates on junctions in the act of disappearing — the opposite: a junction that pulls harder shortens, which makes it pull harder still. So x_e is the edge tension, $\Lambda m_e + 2K_P(P_f - P_f^0) + \Gamma P_f$, taken from the same energy the mechanics minimises.

The two versions cannot be told apart by the spread of junction lengths, because in the fast-myosin limit — the correct one, since myosin turns over in seconds — substituting the target into Eq. (7) turns the length version into a line tension plus a harmonic spring on each edge, and narrowing that spread is what such a spring does by definition. They differ in how often cells swap neighbours, so that is the observable we report: keyed to tension, the rate of neighbour exchange is roughly double.

Two things the model does not have. The myosin responds to the *magnitude* of tension and has no preferred direction in the plane of the sheet, so it cannot express the planar polarity that drives directional tissue elongation. And τ_m is a chosen number, not a measured one.

8. Where this stops, and what it measured

Sixty-six runs (91–164, archived) were spent asking where the sheet sits and what holds it there, and they answer that question and no other. Four results are worth carrying: the standoff is a *tracking lag*, $\delta - \dot{R} \Delta t \gamma / \kappa$, not a force balance, which two fibre lengths differing fourfold confirmed to 3×10^{-5} box units; a force routed through the grid leaves the sheet's deformation gradient carrying 100% of the true stretch while the same force integrated by the engine leaves it carrying 13%; an MPM fibre cannot act as a rope, because $\mathbf{F}$ is advected by the grid's velocity gradient and never measures the distance between two particles, so two ends more than a cell apart are two bodies; and the best configuration reached (153) put the membrane at +0.0011 box units with 2.41 of the 2.43 stretch the geometry implies. What none of them is, is a model derived from what a basement membrane is. The rest of this note starts from that instead.

9. Spheroid $\leftrightarrow$ BM $\leftrightarrow$ ECM

Three objects, and every arrow between them is a measured mechanism rather than a modelling choice. The first thing the measurements say is that the middle object is not a passive spacer. A basement membrane is ~ 100 nm thick, its adhesions are *squat* — plaques ~ 350 nm across on ~ 700 nm centres, joined to the cell across a ~ 30 nm gap [18], so an adhesion is three times wider than the sheet is thick and a third as long — and the sheet is 10 to 100 times stiffer than the stroma outside it, not equal to it. The second thing they say is that it is not a permanent solid: collagen IV in the fly embryo has a half-life of 7–10 hours [17] and in mouse hair follicle about 3, and slowing that turnover disrupts morphogenesis. A sheet enclosing a tissue that triples its radius does not stretch to 2.4; it is *replaced* at the new size while it is being stretched.

Mass, not just mechanics. The sheet carries an areal density ρ of collagen IV that is produced by the epithelium

What this asks of the implementation. Four things, and each is a separate operator with its own test: a mass balance on the sheet (Eq. 8) rather than a fixed particle budget; a strain-dependent modulus (Eq. 9) rather than one number; an adhesion that returns its reaction and permits slip (Eq. 10) rather than a tether to a frozen direction; and an anchorage to the stroma (Eq. 12) that removes the epithelium’s direct push on it.

10. The same thing as a Plexus2 container

Two questions decide whether the biology above can be written down: what the *sets* are, and what kind of operator each mechanism is. Both have answers the literature forces, and both differ from what the archived line built.

Sets. An adhesion is not a body and a fibril is not a particle — each is a *relation between two entities*, which in Plexus2 is an edge-set carrying `.pre` and `.post` index buffers rather than a set of nodes with positions. Writing the hemidesmosome as an edge-set is what makes its reaction automatic: one operator on `plaque` returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. The archived line made the adhesion first a field on the membrane and then a column of MPM particles, and both lost the reaction.

set	kind	one element is	state it carries
spheroid	node, depth 0	the organ	centroid, cell count
cell	node, depth 1	one epithelial cell	A, P, V and their targets, age
vertex	node, depth 1	a corner of the cell mesh	<code>pos, vel</code>
junction	$\text{edge} \subseteq \text{vertex}^2$	a cell–cell contact	length, myosin
bm	node, depth 1	a patch of basement membrane	<code>pos, F, ρ</code> (areal density), age since deposition
plaque	$\text{edge cell} \rightarrow \text{bm}$	one hemidesmosome	rest length ℓ_0 , load, bound flag
ecm	node, depth 1	a patch of interstitial matrix	<code>pos, F</code> , fibre direction
fibril	$\text{edge bm} \rightarrow \text{ecm}$	one anchoring fibril	rest length, load
mpm_grid	field	momentum on a background lattice	mass, momentum
protease	field	MMP concentration	scalar, diffusing
growth_factor	field	FGF/VEGF bound by perlecan	scalar, bound + free

Operators. One row per mechanism in Fig. 3b, in Plexus2’s own taxonomy: *Lateral* moves state along an edge set, *Exchange* scatters to or gathers from a field, *Rewire* changes which elements are connected, and *Divide/Die* change how many elements exist. The last column is what would falsify the operator — the measurement its test folder has to make.

operator	kind	acts on	mechanism, from the literature	what its test measures
<i>epithelium</i>				
grow_3d, divide_3d	Divide	cell	volume growth and cytokinesis [3, 4]	cells 200 $\rightarrow$ 6076, radius $\times 3.56$
shape_energy_3d	Lateral	junction	vertex-model force balance [5]	aspect 1.021, stays round
reconnect_t1_3d	Rewire	junction	neighbour exchange [4]	0.0089 T1 per cell per frame
<i>epithelium $\rightarrow$ basement membrane</i>				
bm_secrete	Divide	bm	polarised deposition of laminin, collagen IV, nidogen, perlecan; the sheet turns over on a half-life of hours [17, 15]	areal density ρ held constant as $A(t)$ grows
bm_assemble	Rewire	bm	laminin polymerises where integrin and dystroglycan nucleate it [15]	no sheet forms where the receptor is absent
bm_degrade	Die	bm	MMP proteolysis, local and MMP-independent breach [20]	a hole opens only where protease is present
<i>basement membrane $\leftrightarrow$ epithelium</i>				
plaque_seed	Rewire	plaque	hemidesmosomes form at density Σ^{-2} , $\Sigma \approx 7T$ [18]	plaque count and spacing
plaque_pull	Lateral	plaque	Eq. (10): normal spring at rest length ℓ_0 , tangential friction ξ , and its reaction [18]	standoff = ℓ_0 ; momentum conserved to machine precision
plaque_rupture	Rewire	plaque	the bond fails past a load	fraction bound vs. applied stress
bm_stiffness	Lateral	bm	Eq. (9): tangent modulus rising with stretch [16, 13]	E 0.4 $\rightarrow$ 3 MPa across the run
<i>basement membrane $\leftrightarrow$ matrix</i>				
fibril_seed, fibril_pull	Rewire, Lateral	fibril	collagen VII anchoring [19]; the stroma is loaded <i>through</i> the sheet	stroma displacement with the sheet present vs. absent
mpm_scatter/gather	Exchange	bm, ecm	both bodies share one momentum field [1, 2]	contact without an explicit contact law
bm_barrier	Exchange	bm, fields	pores 10–130 nm pass solutes, stop cells [15]	solute flux across an intact vs. breached sheet
<i>matrix</i>				
ecm_align, ecm_crosslink	Lateral	ecm	fibres rotate toward the maximum principal stretch, which is what carries force beyond one cell diameter [11, 12]	alignment angle vs. radial direction

Three of these have no counterpart in the archived implementation and are the reason it stalled: **bm_secrete** as a mass balance rather than a fixed particle budget, **plaque_pull** returning its reaction, and **fibril_pull** replacing the epithelium’s direct push on the stroma. Each gets its own folder, spec, movie and measurement plot, in the order of the table.

11. Symbols

Grouped by what they belong to. Where a symbol is a stiffness, the second column says what it is a stiffness *of*, since several are easy to confuse.

Epithelium — Eqs. (1), (2)		
A_f, P_f, V_f	area, perimeter, volume	of cell (face) f ; A_f^0, P_f^0, V_f^0 are its targets
K_A, K_P, K_V	stiffnesses	how hard a cell resists changing its area / perimeter / volume
Λ	line tension	pull along every cell–cell contact, one value for the whole tissue
Γ	perimeter contractility	a second, quadratic pull on each cell’s perimeter
ℓ_e	length of junction e	
K_R, R_0	radial restoring stiffness, radius	holds the shell against collapse
f_{grow}	per-frame growth factor	applied to a cell’s target volume
Junction myosin — Eq. (7)		
m_e	myosin on junction e	dimensionless multiplier on Λ ; mean 1
x_e	what the feedback reads	the junction’s tension
a	global myosin level	multiplies every junction equally
β	feedback strength	how strongly m_e responds to x_e above average
τ_m	myosin timescale	how fast m_e chases its target
Surface and matrix — Eqs. (3), (4)		
$\hat{\mathbf{u}}_i$	direction	fixed unit vector from the tissue centre $\mathbf{c}$
$R(\hat{\mathbf{u}}, t)$	surface radius	how far the epithelium reached in that direction
r_i	radius of particle i	
k_c	contact stiffness	how hard the surface pushes a fibre back out
P	matrix pressure	contact force per unit surface area, by direction
$p_{1/2}, n, f$	gate half-point, sharpness, floor	g is the resulting growth multiplier, $f \leq g \leq 1$
Basement membrane — Eqs. (5), (6)		
$\mathbf{x}_i$	position of node i	
γ	drag	converts force to speed: $\dot{\mathbf{x}} = \mathbf{F}/\gamma$
k_b	crosslink stiffness	how hard a stretched crosslink pulls its two nodes together
L_{ij}, ℓ_{ij}^0	current, rest length	of the crosslink between nodes i and j
$\hat{\mathbf{d}}_{ij}$	unit vector	from node i toward node j
τ_r	remodelling timescale	how fast ℓ^0 creeps toward L , i.e. how fast it forgets
ε_b	break strain	the stretch at which a crosslink snaps
κ	tether stiffness	how hard a node is pulled back to its anchor (not k_b)
$\mathbf{a}_i$	anchor	the point on the surface node i is tied to
δ	offset	how far outside the surface the sheet sits

12. References

The operators of §10 are grouped here by what they claim. Where a row of that table states a number — a half-life, a plaque spacing, a modulus — the citation is the measurement it was taken from, not a review that mentions it.

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